THE PULLBACK OF THE QUASI-COHERENT MODULE ASSOCIATED TO A MODULE IS ASSOCIATED TO THE BASE CHANGED MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the quasi-coherent module associated to a module as the pullback along the structure morphism of the one point space. Invoked inside the counted proof at source lines 1448-1450 as a pointer to a lemma of this chapter. Reduces the statement to the compatibility of pullbacks with composition.
- The commutativity of the square of ringed spaces formed by the structure morphisms and the given morphism. Written down in the counted proof (source lines 1452-1460), with a cross-chapter pointer for the composition of pullbacks. Identifies the two pullbacks of the module along the two routes of the square.

Source reference. Original lemma lemma-construct-quasi-coherent-sheaves: sources/stacks-modules/pinned-original-source.tex, source lines 1425–1425. Navigation pointer only: the counted text refers to the Stacks item lemma-construct-quasi-coherent-sheaves, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference sheaves-lemma-push-pull-composition-modules: sources/stacks-modules/pinned-original-source.tex, source lines 985–985. Navigation pointer only: the counted text refers to the Stacks item sheaves-lemma-push-pull-composition-modules, which belongs to another chapter of the project and is

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Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Set $R = \Gamma(X, \mathcal{O}_X)$. Let M be an R-module. Let $\mathcal{F}_M$ be the quasi-coherent sheaf of $\mathcal{O}_X$-modules associated to M. If $g : (Y, \mathcal{O}_Y) \to (X, \mathcal{O}_X)$ is a morphism of ringed spaces, then $g^*\mathcal{F}_M$ is the sheaf associated to the $\Gamma(Y, \mathcal{O}_Y)$-module $\Gamma(Y, \mathcal{O}_Y) \otimes_R M$.*

Proof. The assertion follows from the first description of $\mathcal{F}_M$ in Lemma lemma-construct-quasi-coherent-sheaves as π^*M, and the following commutative diagram of ringed spaces

$$\begin{array}{ccc} (Y, \mathcal{O}_Y) & \xrightarrow{\pi} & (\{*\}, \Gamma(Y, \mathcal{O}_Y)) \\ \Big\downarrow{\scriptstyle g} & & \Big\downarrow{\scriptstyle \text{induced by } g^\sharp} \\ (X, \mathcal{O}_X) & \xrightarrow{\pi} & (\{*\}, \Gamma(X, \mathcal{O}_X)) \end{array}$$

(Also use Sheaves, Lemma sheaves-lemma-push-pull-composition-modules.) □